Project Proposal

PRANA

Perception-conditioned Robotic Action Network with Attention

A Custom Vision-Language-Action Policy for Robotic Manipulation

Author	Siddarth Dayasagar, Arpit Gandhi
Program	M.S. Robotics (ECE), Northeastern University
Course	Pattern Recognition and Computer Vision
Date	March 2026
Repository	github.com/siddarth09/PRANA

1. Abstract

This proposal introduces PRANA (Perception-conditioned Robotic Action Network with Attention), a custom Vision-Language-Action (VLA) policy designed to generate temporally consistent robot actions from multimodal perception. PRANA fuses visual observations with natural language task instructions through a transformer-based architecture, producing smooth, executable action trajectories for robotic manipulation.

The project will be executed in two phases: building the complete PyTorch pipeline from scratch, followed by integration into the Hugging Face LeRobot framework for hardware deployment. The initial demonstration task is autonomous screwdriver pick-and-place using a LeRobot hardware setup.

2. Motivation

Vision-Language-Action models represent a rapidly growing area at the intersection of computer vision, natural language processing, and robotics. Existing VLA systems such as OpenVLA and RT-2 demonstrate that end-to-end transformer architectures can map visual observations and language commands directly to robot actions, but these models are large, opaque, and difficult to customize for specific hardware constraints.

Building PRANA from the ground up serves two purposes:

- Deep understanding of how attention mechanisms, tokenization strategies, and action decoding interact within a multimodal transformer pipeline.
- A practical foundation for the Intrinsic "AI for Industry" challenge, where a UR5 arm must perform a dexterous cable-insertion task. Developing PRANA on LeRobot hardware first establishes the necessary intuition for handling action jitter, temporal consistency, and sim-to-real transfer before scaling to a more complex platform.

3. Technical Approach

3.1 Architecture Overview

PRANA processes three input streams through specialized encoders before fusing them in a shared transformer backbone:

Component	Method	Details
Vision Encoder	Pretrained ViT (DINOv2 / SigLIP)	Extracts spatial feature tokens from RGB observations; frozen or fine-tuned
Language Encoder	Pretrained text transformer	Encodes natural language task instructions into token embeddings
Proprioceptive Input	MLP projection	Projects joint positions and gripper state into the shared embedding space

Fusion Backbone	Causal Transformer	Cross-attends over vision, language, and proprioceptive tokens to produce a fused representation
Action Head	Chunked action decoder	Predicts a sequence (chunk) of future actions per forward pass for temporal smoothness

3.2 Action Chunking

Rather than predicting a single next-step action, PRANA outputs a chunk of K future actions per inference step. This design choice directly addresses action jitter, a common failure mode in frame-by-frame policies. Overlapping chunks are blended using exponential weighting to ensure smooth transitions.

3.3 Training Objective

The model is trained end-to-end with a mean squared error (MSE) loss between predicted and ground-truth action chunks collected from teleoperated demonstrations. An optional auxiliary loss on predicted end-effector pose may be added to improve spatial grounding.

4. Data Collection

Training data will be collected through teleoperated demonstrations on the LeRobot hardware platform. Each demonstration captures:

- RGB image stream from the workspace camera
- Joint position and gripper state at each timestep
- A natural language instruction describing the task (e.g., "Pick up the screwdriver and place it in the box")

For the initial milestone, we target approximately 50 to 100 successful demonstrations of the screwdriver pick-and-place task. Data will be stored in the LeRobot dataset format for compatibility with Phase 2 integration.

5. Project Phases

Phase 1: From-Scratch PyTorch Implementation

The goal of Phase 1 is to build every component of PRANA manually to ensure a thorough understanding of the architecture:

Milestone	Deliverable	Target
M1	Dataset pipeline: dataloader, tokenizer, and augmentation	Week 1-2
M2	Vision encoder integration (ViT with patch embedding)	Week 2-3

M3	Language encoder and cross-attention fusion module	Week 3-4
M4	Action chunking decoder and training loop	Week 4-5
M5	End-to-end training on collected demonstrations	Week 5-6
M6	Evaluation: success rate, trajectory smoothness, jitter metrics	Week 6-7

Phase 2: LeRobot Integration and Deployment

Phase 2 ports the trained architecture into the Hugging Face LeRobot library, leveraging their standardized training, evaluation, and deployment pipeline:

- Register PRANA as a custom policy within LeRobot's configuration system.
- Validate that training and inference produce equivalent results to the Phase 1 standalone pipeline.
- Deploy the model on the physical LeRobot hardware for real-world pick-and-place execution.
- Profile inference latency and optimize for real-time control loop requirements.

6. Evaluation Criteria

The project will be evaluated across both quantitative and qualitative dimensions:

Metric	Description
Task Success Rate	Percentage of trials where the screwdriver is successfully picked and placed in the box
Action Smoothness	L2 norm of action deltas across consecutive timesteps; lower indicates less jitter
Temporal Consistency	Variance of predicted action chunks across overlapping windows
Inference Latency	Time per forward pass; must meet real-time control loop requirements (target < 50ms)
Generalization	Qualitative evaluation on slight variations in object position and lighting

7. Tools and Resources

- Hardware: LeRobot manipulation platform with workspace camera
- Frameworks: PyTorch, Hugging Face Transformers, Hugging Face LeRobot
- Pretrained Models: DINOv2 or SigLIP (vision), lightweight text encoder (language)
- Compute: Local GPU workstation and/or cloud instances for training

8. Broader Impact and Future Work

PRANA is designed with extensibility in mind. The architecture and training pipeline built during this project will directly transfer to the Intrinsic "AI for Industry" challenge, where the task complexity increases to dexterous cable insertion on a UR5 industrial arm. Key future extensions include:

- Scaling to multi-task training with diverse manipulation primitives.
- Incorporating depth and tactile sensing modalities.
- Exploring diffusion-based action heads as an alternative to deterministic chunked decoding.
- Sim-to-real transfer using domain randomization in simulation environments.

10. References

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